AWS to Sophos Site-to-Site VPN

Configuration, troubleshooting and verification record

VPN connection	vpn-029f060b34b9a0b26 (eatngo-csu-aws-vpn)
Routing / IKE	Static routing, IKEv2, pre-shared key
Virtual private gateway	vgw-02b88e67cc7fef962
Customer gateway	cgw-0f06b9a6fa3ddaeee (HQ Sophos Firewall, public IP 197.234.39.74)
AWS VPC	vpc-0c345c86a2c920a60 (Default VPC, 172.31.0.0/16)
HQ networks	172.16.0.0/16 and 192.168.0.0/24
Tunnels	Tunnel 1 configured and up. Tunnel 2 not configured (test deployment).
Date	21 October 2026

1. Summary

A site-to-site IPsec VPN was built between an AWS Virtual Private Gateway and the HQ Sophos Firewall so that HQ networks and the AWS VPC can reach each other. The Sophos side uses a route-based tunnel interface (xfrm78). The tunnel came up first; ping then failed until several issues were fixed (see section 6). The last fix was a typo in an AWS static route (172.0.0.0/16 instead of 172.16.0.0/16). After that, ping worked in both directions between the EC2 test instance and HQ hosts.

2. Addressing

Item	Tunnel 1	Tunnel 2 (not configured)
AWS outside IP (VGW)	54.173.139.134	54.175.215.223
AWS inside IP	169.254.23.189/30	169.254.221.221/30
Sophos inside IP	169.254.23.190/30	169.254.221.222/30
Sophos outside IP (CGW)	197.234.39.74	197.234.39.74
Sophos tunnel interface	xfrm78 (connection AWSTunnel1)	-

Pre-shared keys are deliberately left out of this document. Store them in a password manager.

3. AWS configuration

3.1 Attach the VGW to the VPC

Route propagation showed no virtual private gateways until the VGW was attached. *VPC* > *Virtual private gateways* > *Actions* > *Attach to VPC* (vpc-0c345c86a2c920a60). A VGW attaches to only one VPC at a time.

3.2 VPN connection settings

Setting	Value
Target gateway	Virtual private gateway vgw-02b88e67cc7fef962
Customer gateway	Existing, cgw-0f06b9a6fa3ddaeee

Setting	Value
Routing options	Static
Static IP prefixes	192.168.0.0/24 and 172.16.0.0/16
Local IPv4 network CIDR	0.0.0.0/0 (default)
Remote IPv4 network CIDR	172.31.0.0/16
IKE version	ikev2
Phase 1 / Phase 2 DH group	14 / 14
Phase 1 lifetime / Phase 2 lifetime	28800 s / 3600 s
Rekey margin / fuzz	270 s / 100 %
DPD timeout / action	30 s / Clear

Only one CIDR belongs in the Local IPv4 network CIDR field, and several traffic selectors allow only one active security association at a time. Leaving both CIDR fields at 0.0.0.0/0 and listing the networks under Static IP prefixes avoids this.

3.3 Route table

On the Main route table (rtb-01dc09099fab71ae4) enable *Route propagation* for the VGW. The Routes tab should then show 192.168.0.0/24 and 172.16.0.0/16 with target vgw-02b88e67cc7fef962 and Propagated = Yes.

3.4 Security group

The EC2 security group allows inbound ICMP. It was set to 0.0.0.0/0 for testing and should be narrowed to the HQ CIDRs.

4. Sophos configuration

4.1 IPsec profile

Configure > *Site-to-site VPN* > *IPsec* > *IPsec profiles* > *Add.* Key exchange IKEv2; Phase 1 and Phase 2 encryption and authentication matching the AWS tunnel options; DH group 14 and PFS group 14; key life 28800 (Phase 1) and 3600 (Phase 2); DPD on with re-initiate.

4.2 IPsec connection

Setting	Value
Name	AWSTunnel1
Connection type	Tunnel interface
Gateway type	Initiate the connection
Authentication	Preshared key (Tunnel 1 key from the AWS configuration file)
Gateway address	54.173.139.134
Listening interface	WAN port holding 197.234.39.74
Local ID / Remote ID	IP address 197.234.39.74 / IP address 54.173.139.134

AWS does not list Sophos in its vendor drop-down. The configuration file was downloaded with Vendor = Generic, Platform = Generic, Software = Vendor Agnostic, IKE version = ikev2.

4.3 Tunnel interface

Network > Interfaces > xfrm78: IPv4 address **169.254.23.190** with netmask **/30** (255.255.255.252). This only configures the tunnel interface in the VPN zone and does not affect the WAN or default route.

4.4 Static route to AWS

Configure > Routing > Static routes > Add:

Field	Value
Destination	172.31.0.0 / 16
Gateway IP	(blank)
Interface	xfrm78-169.254.23.190
Administrative distance / metric	1 / 0

Sophos rejects a link-local address (169.254.x.x) as a Gateway IP, so the route is created on the interface with the gateway left empty. It covers only 172.31.0.0/16, so it does not affect internet traffic.

4.5 Firewall rules

Protect > Rules and policies: rule LAN-VPN (rule ID 6) accepts traffic from the HQ networks to 172.31.0.0/16, with no NAT applied. A matching VPN to LAN rule allows replies and AWS-initiated traffic. Masquerading must stay off for these rules so AWS sees the real HQ source addresses.

5. Verification

Test	Result
AWS console, Tunnel 1 status	Up
Sophos Route lookup for 172.31.26.21	Located on xfrm78
Sophos Log viewer, ICMP 172.16.0.236 to 172.31.26.21	Allowed by rule LAN-VPN (6), NAT rule 0, out interface xfrm78
EC2 (172.31.26.21) to 192.168.0.2 and 172.16.0.2	Replies received
HQ to EC2 and EC2 to HQ	Ping works in both directions
AWS Tunnel 2 status	Down (not configured, expected)

6. Issues found and how they were resolved

#	Symptom	Cause	Resolution
1	Sophos missing from the vendor list when downloading the config	AWS has no Sophos entry in the drop-down	Download as Generic / Generic / Vendor Agnostic / ikev2
2	Route propagation tab: no virtual private gateways	VGW was not attached to the VPC	Attach vgw-02b88e67cc7fef962 to the Default VPC, then tick it under Route propagation
3	Static route rejected with 'cannot enter a link-local IP for Gateway IP'	Sophos does not accept 169.254.x.x as a gateway	Leave Gateway IP blank and select the xfrm78 interface

#	Symptom	Cause	Resolution
4	Tunnel up, but ping failed from both sides to 172.16.0.0/16 hosts	AWS static route entered as 172.0.0.0/16 instead of 172.16.0.0/16	Remove 172.0.0.0/16, add 172.16.0.0/16, confirm the propagated route
5	Tunnel 2 shown as Down	Never configured on Sophos; AWS does not initiate by default	Not needed for the test. Configure later if redundancy is required.

The Sophos firewall log and the AWS security group were correct throughout. Sophos showed the packet allowed and sent out xfrm78, which pointed the investigation at the AWS side.

7. Extending the VPN to a store (in progress)

A separate store tunnel was added to the existing HQ connection so AWS can reach the store LAN through HQ. Phase 2 for the AWS subnet and the store subnet shows established. Two items are still needed for traffic to flow:

- Add the store LAN subnet to the AWS VPN connection's static routes (unless covered by 192.168.0.0/24 or 172.16.0.0/16), and confirm it shows as propagated in the route table.
- On Sophos, create VPN to VPN firewall rules in both directions between 172.31.0.0/16 and the store subnet, with masquerading off. On the store router, allow the traffic in the forward chain and add a srcnat accept rule above the masquerade rules.
- Note: if a store already has its own tunnel to an AWS network using 172.31.0.0/16, the two networks overlap and traffic cannot be routed to both. A test VPC with a non-overlapping CIDR (for example 10.60.0.0/16) avoids this.

8. Security and cleanup

Done	Task
[ ]	Rotate both pre-shared keys in AWS (Tunnel options) and update Sophos. The keys were shared outside the password manager.
[ ]	Restrict the EC2 security group ICMP rule from 0.0.0.0/0 to the HQ CIDRs, or remove it after testing.
[ ]	Back up the Sophos configuration (Backup and firmware > Backup).
[ ]	Terminate any leftover test EC2 instances and release unused Elastic IPs.
[ ]	Configure Tunnel 2 with a higher administrative distance (for example 5) when redundancy is needed.

9. Troubleshooting quick reference

```
# Sophos advanced shell
tcpdump -ni xfrm78 icmp

# Sophos GUI
Diagnostics > Tools > Route lookup        (confirm 172.31.x.x goes out xfrm78)
Log viewer > Firewall                     (shows the rule ID that allowed or dropped)
Log viewer > IPsec                        (negotiation errors)

# AWS console
VPC > Site-to-Site VPN > Tunnel details   (tunnel state)
VPC > Site-to-Site VPN > Static routes    (prefixes must be correct and Available)
VPC > Route tables > Routes               (propagated routes via the VGW)
```